

Surgical Knowledge Swapping in Large Language Models

Project Overview

This project investigates precision model editing in LLMs using ROME, MEMIT, and LoRA. We evaluate edit success, generalization, locality, consistency, and interference.

Objectives

- Localize knowledge in transformers
- Apply targeted edits
- Compare editing methods
- Evaluate robustness

Evaluation Metrics

ESR, PG, LL, CS, IR

Repository Structure

```
project/  
  data/  
  src/  
  scripts/  
  outputs/  
  reports/  
  README.md
```

Setup

```
git clone <repo>  
conda create -n nlp-edit python=3.10  
pip install -r requirements.txt
```

Pipeline

Load model → prepare data → causal tracing → apply edits → evaluate → analyze

Timeline

Week 1: Setup

Week 2: Experiments

Week 3: Analysis

Team Roles

Each member owns one module: tracing, ROME, MEMIT, LoRA, evaluation

Deliverables

Code, results, report